# Universal gravitational energy

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use the zero-at-infinity convention for gravitational potential energy.

**SUCCESS CRITERIA - I can...**

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “potential energy” and “radius” to explain your model or result. Sentence starters are optional.

**potential energy**: energy associated with the configuration of interacting objects

**radius**: distance from the center to the edge of a circle or sphere

**initial**: at the beginning of the interval; subscript i

## 01 Bring back an earlier idea

A cart’s work is measured over the first 3 m, but its speed change is measured over 5 m. Can these values directly test $W_{net}=\Delta K$?
**Explain**.

**Word help:** work = energy transferred by a force along a displacement

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; choose U = 0 at infinity. Moving outward makes U less negative.

**Worked reasoning:** For a supplied source with GMm=120 J·m, potential energy is U=-120/r. At r=3 m, U=-40 J; at 6 m, U=-20 J. Moving outward raises U by 20 J.

**Your turn: Draw** an energy number line and mark the direction toward the zero at infinite separation.

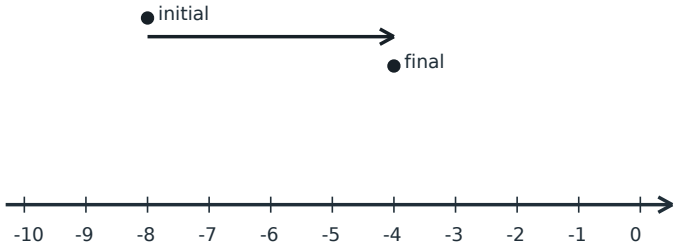
**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

An object moves outward from r to 2 r. Does gravitational potential energy become more or less negative?
**Draw** a number-line comparison.

**Sentence starter:** The … in my model represents … . Its label shows … .




Example: potential energy (J)

**Marzano check - circle:** ① ② ③

## 04 Build the solution

Given GMm=24 J·m, find U at r=3 m and 6 m, then ΔU for moving outward.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

A student says “negative potential energy means the object has negative kinetic energy.” Correct the claim.

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

In the same straight-line deflection model, keep sideways Δv fixed but make lead time 4 times as long.
**Compare** shift and name a limitation.

**Word help:** deflection = a change in the direction or path of motion

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

U changes from -10 J to -6 J. CalculateΔU and state whether the object gained gravitational energy.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Return to the worked example in problem 2.
**Use** the same numerator to find U at 4 and 12 m. Is the energy change positive or negative when moving outward?

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3